Jialin Huang

Software Engineer

linao264590000@gmail.com github.com/jialinhuang00 linkedin.com/in/jlhuangprogramming jialin00.com

Open to relocation anywhere · Mandarin (native), English (professional)

Software engineer, eight years. Most recently on a multi-tenant security SaaS, writing Go and Node services for identity and access: claims-based RBAC on Cognito and GraphQL, SAML and social sign-in, and a Go audit pipeline for auth events. On the build side I moved the monorepo to pnpm (installs 3-10x faster) and cut two Docker images by about 80%. Frontend background; React and TypeScript daily.

Experience

MammothCyber Senior Software Engineer
Apr 2025 - Jul 2026
Cybersecurity · Taipei (Hybrid)
Full-stack across the web portal of a multi-tenant enterprise security platform, a Chromium-based secure browser, and Go services; identity & access domain (SAML, OIDC/JWT, RBAC, audit, policy config)

Identity, Access & Audit

- Worked across the auth stack (Go + Node): added tenant-level forced re-authentication (SAML), and threaded an OS / device-posture field across the login-policy service, the Cognito audit pipeline, and the browser and extension clients
- Claims-based RBAC end to end: permissions computed in the Cognito PreTokenGeneration Lambda and compressed into the id_token, enforced by a Casbin guard behind a GraphQL @rbac directive
- Auth-event audit pipeline in Go: Cognito CloudTrail events polled from SQS, classified (password / MFA / device failures), enriched from PostgreSQL with an LRU cache, shipped to Logstash
- Added Google / Apple social login (OAuth) to Cognito alongside the existing SAML, reworking the Lambda triggers that map each social user to its tenant

Platform & Build Infrastructure

- Drove a monorepo dependency cleanup on npm: root overrides and targeted upgrades (Express v5, Apollo Server v5, ESLint v9) cut vulnerabilities 74 → 13 (-82%), clearing every high & critical, and stray local packages 22 → 0
- **npm to pnpm**: Migrated the monorepo to pnpm workspaces with a staged rollout (shamefully-hoist bridge, declare every phantom dep, flip to strict isolation), cutting installs 3-10x and making per-worktree installs cheap so parallel AI agents can each build in their own git worktree
- Cut two Docker images ~80% in three steps: per-workspace pnpm deploy --prod instead of copying the root node_modules, misdeclared prod deps fixed, then both servers bundled (Remix ssr.noExternal + esbuild, NestJS nest build --webpack) so the runtime image ships no node_modules. Remix web app 1.88 GB → 264 MB, NestJS API 1.95 GB → 490 MB
- Wrote the integration tests in Playwright, running pages in parallel

Policy Configuration Console

- Built the policy-configuration backend in NestJS: rule and content-profile APIs, RBAC-gated, with audit logging and an N+1 fix that collapsed a multi-call save into a single request
- Built the rules and content-profile editor in Remix, type-safe end to end with reactive validation

Secure Browser & Extension (Chromium)

- Shipped a login / session-visibility feature across every client of a Chromium-based enterprise browser, extending each platform's native bridge in C++ (Android / Desktop) and Objective-C++ (iOS), plus a Go fetch backend and a React UI

Console Migration to Remix

- Migrated console pages onto Remix (React 18, Mantine, TanStack Query/Table, Vite) off an earlier Vue / Next.js stack, page by page with new and old running side by side

TypeScript · NestJS · Apollo GraphQL · React · Next.js · Remix · Vue · Golang · Gin · PostgreSQL · Redis · Elasticsearch · Docker · Kubernetes · Jenkins · AWS Lambda · AWS Cognito · Playwright

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Bunastar Frontend Engineer

Nov 2024 - Apr 2025

Gaming · Taipei (Hybrid)

Real-time multiplayer card game for the Southeast Asian market; custom Canvas engine and WebSocket sync

Southeast Asian Gaming Platform - Pok Deng

- Card game logic: dealing, turn order, scoring, round settlement
- Real-time card sync over WebSocket (client side)
- Canvas rendering engine, optimized with sprite sheets
- Cross-domain communication via postMessage API

TypeScript · React · Canvas · WebSocket · Google Analytics

Confetti Full Stack Engineer

Dec 2022 - Oct 2024

Gaming · Taipei (Hybrid)

confetti.studio ↗

- Y2K landing page

Gaming Website Rewrite

- Rewrote the platform API in Node.js
- Rewrote the website in Next.js, React and Tailwind CSS

TypeScript · React · Next.js · Tailwind CSS · Node.js

SCMP Frontend Engineer

Oct 2021 - Nov 2022

Media · Hong Kong (Remote)

South China Morning Post: high-traffic international news platform, delivered fully remote

scmp.com ↗

- Improved Core Web Vitals (LCP, CLS), Google's SEO ranking signals, on scmp.com, a high-traffic international news site: fixed the homepage hero's layout shift, lazy-loaded below-fold images, and tuned SSR data prefetch
- Applied GDPR right-to-erasure in the comment system: surfaced a gdpr_erased flag through the GraphQL layer so erased users' names render as Anonymous
- Maintained the CMS-driven content rendering pipeline (Node.js, GraphQL)
- Built the site-wide tracking plan across sections: GA events through the GTM dataLayer for traffic, and Mixpanel via SDK keyed on the reader's uuid for per-user product behavior

TypeScript · Vue · Next.js · React · Node.js · GraphQL

Futuresai Frontend Engineer

Nov 2020 - Oct 2021

Finance · Taipei (Hybrid)

Trading Dashboard + Admin + Auth

- Monorepo rewrite with SignalR for real-time trading-data streaming
- Firebase auth middleware with custom membership logic, plus SMS and LINE push notifications

futures-ai.com ↗

- Built a subscription-billing system with multiple tiers
- Live trading strategy articles with graphical analysis

TypeScript · React · Next.js · Emotion · SignalR · Firebase Auth · Firebase

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Dentalltw Frontend Engineer

Jul 2018 - Nov 2020

Healthcare · Taipei (Hybrid)

Built a 0-to-1 clinic product: a desktop app for Taiwan and a React web app for the Vietnam market

Dentall Clinic Management ↗

- Built the desktop clinic app (UWP, C#/.NET) for registration, appointments, billing, and prescriptions, integrated with Taiwan's NHI (National Health Insurance) system
- Built an SVG-based interactive dental chart for procedure documentation

Vietnam Market Expansion

- Built the Vietnam web app (React, Redux): the clinic software localized, with multi-language support and SMS

CI/CD Pipeline

- Set up CI/CD to build and push Docker images to GCR, using a Go service and GitLab scripts

TypeScript · React · Redux · C#/.NET · Python/Flask · Docker · GCP

Skills

Languages	TypeScript, Go
Backend	NestJS, Apollo GraphQL, Gin, Elasticsearch, Node.js, GraphQL, SignalR, Firebase Auth, C#/.NET, Python/Flask
Security & Domain	RBAC, SAML, OAuth2/OIDC, token claims, audit logging, Zero Trust access, multi-tenant SaaS
Infrastructure	PostgreSQL, Redis, Docker, Kubernetes, Jenkins, AWS Lambda, AWS Cognito, Playwright, Firebase, GCP
Frontend	React, Next.js, Remix, Vue, Canvas, WebSocket, Google Analytics, Tailwind CSS, Emotion, Redux

Projects

Palimpsest ↗

A boilerplate for running an AI coding agent under enforced rules: an identity file models the user, and shell hooks refuse any write that breaks a rule, instead of asking the model to comply.

- Three governance speeds in one charter: a bottomline no in-session pressure can move, principles that shift only on new evidence, and a free drift layer.
- An Angular dashboard comparing 11 open-source "AI identity" projects, built by cloning each repo and reading its config: 5 of the 11 enforce their constraints, the rest only declare them.

Angular · TypeScript · Bash · Claude Code

URL Shortener ↗

One redirect touches five systems: PostgreSQL for link mappings, Redis for live counters, Kafka for the click pipeline, ClickHouse for analytics, and a Go/Gin service tying them together.

- Clicks flow through Kafka in 500-event / 2s batches across 4 writer goroutines, so analytics writes never block a redirect
- Write-behind counter: redirects INCR in Redis, a 5s worker drains to PostgreSQL; reads merge the durable total with the live delta, streamed to a dashboard over Server-Sent Events

Go · Gin · PostgreSQL · Redis · Kafka · ClickHouse

221b

RAG over my own writing: blog posts, AI-agent sessions, and config files in one corpus, queried in Chinese, answered in my English voice.

- Chinese question → multilingual-e5 retrieval over ~4,000 chunks → bge cross-encoder rerank → answer grounded in my English writing.
- Eval harness reports Recall@5 and MRR against ground-truth articles, so a model or prompt swap shows up as a number.

Python · ChromaDB · sentence-transformers · MCP

Game of Life ↗

Conway's Game of Life running on the GPU.

- Simulated 770K+ cells at 60fps with a WebGL fragment-shader pipeline

WebGL · TypeScript

Kubelens ↗

If you have kubectl, you have Kubelens. Nothing to install.

- Interactive graph of how Deployments, Pods, Services, and Ingresses connect; drag any resource into its own panel and run rollouts there.
- Snapshot the whole cluster to browse offline.

Angular · TypeScript

Writing

- N+1 isn't a database problem. It's a shape. ↗
- HTTP 1.1, 2, and 3: what broke and what fixed it ↗
- Five ways to reach a service in Kubernetes ↗
- A packet's walk through a VPC, in AWS ↗
- Stopping XSS and CSRF: who's responsible for what ↗
- Inside React's FiberWorkLoop ↗
- CSS rendering: layers, threads, and why transform beats width ↗
- from setState to pixels ↗
- SPA is not CSR: the 2026 rendering strategy map ↗
- Hiroki's Ark ↗

Education

National Chengchi University (NCCU) B.A. in Economics